

Sequences and Series

ANSWER KEY



Sequence limits

- 1 Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$ Converges to 3 (divide by n).

b $a_n = \frac{n^2}{2^n}$ Converges to 0 (exponential growth dominates polynomial).

c $a_n = (-1)^n$ Diverges (oscillates between -1 and 1).

- 2 Evaluate $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$.

This is the standard limit form giving e^3 .

Series and partial sums

- 3 Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.
Diverges — this is the harmonic series, a standard example of a divergent series with terms going to 0.

- 4 Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.
 $|r| = \frac{2}{5} < 1$: $S = \frac{3}{1-2/5} = 5$.

- 5 Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.
 $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. The partial sums telescope: $S_N = 1 - \frac{1}{N+1} \rightarrow 1$.

Monotonic and bounded sequences

- 6 Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n .
 $a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$, so $a_{n+1} > a_n$: the sequence is increasing.

- 7 Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.

By the Monotonic Convergence Theorem, a bounded, monotonic sequence converges. $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$ (divide numerator and denominator by n).

More series practice

- 8 Evaluate the series $\sum_{n=0}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n \right]$ by splitting into two geometric series.
 $\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{1-1/2} = 2$. $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n = \frac{1}{1-1/3} = \frac{3}{2}$. Total = $2 + \frac{3}{2} = \frac{7}{2}$.

- 9 A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.
- $$\text{Total} = 10 + 2(10(0.6) + 10(0.6)^2 + \dots) = 10 + 2 \cdot \frac{6}{1-0.6} = 10 + 2(15) = 40 \text{ m.}$$
-

Recursive sequences

- 10 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2 , a_3 , and a_4 .
- $$a_2 = \frac{2+6}{2} = 4. \quad a_3 = \frac{4+6}{2} = 5. \quad a_4 = \frac{5+6}{2} = 5.5.$$
- 11 Assuming the recursive sequence above converges to a limit L , find L by setting $L = \frac{L+6}{2}$ and solving.
- $$2L = L + 6, \text{ so } L = 6.$$
-

Repeating decimals as series

- 12 Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.
- $$S = \frac{\frac{45}{100}}{1 - \frac{1}{100}} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}.$$